Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	*(a)*			$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ (1) Any of the following for (1) • During this process the iron(III) ions (in Fe_2O_3) gain electrons (to produce iron); reduction is a process of electron gain • The oxidation number of iron is reduced from +3 to 0; a reduction in (positive) oxidation number is reduction • Carbon monoxide loses electrons; oxidation is a process of electron loss • The oxidation number of carbon is increased from +2 to +4; an increase in (positive) oxidation number is oxidation • Fe_2O_3 loses oxygen and CO gains oxygen	2			2		
	(b)			350 tonnes of which 0.02 % is sulfur $\therefore$ Mass of sulfur $= \dfrac{350 \times 0.02}{100} = 0.07$ tonnes (1) $Mg + S \rightarrow MgS$ 0.07 tonnes sulfur needs $\dfrac{24.3 \times 0.07}{32.1} = 0.0530$ tonnes $= 53.0$ (kg) (1) ecf possible	2			2	1	
	(c)			Cubic structure shows alternating different ions (1) Ions labelled as Mg^{2+} and S^{2-} (1)		2		2		

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					AO1	AO2	AO3	Total	Maths	Prac
8.	*(d)*	(i)		White solid / precipitate (of magnesium hydroxide)		1		1		1
		(ii)		Number of mol of MgS $= \frac{0.224}{56.4} = 0.00397$ $\therefore$ 0.00397 mol of H_2S also produced (1) Volume of $H_2S = 0.00397 \times 24.0 = 0.095(3)\ dm^3$ (1) $= 95.(3)\ cm^3$ (1) ecf possible Accept alternative method using $pV = nRT$	1	2		3	1	
	(e)			A colourless solution (1) The solubility of the group 2 hydroxides increases down the group (1)	1	1		2		1
	(f)			$BaO + H_2O \rightarrow Ba(OH)_2$ (1) pH > 7 (1)	2			2		
	(g)			Any two of following for (1) each • Barium is in group 2 and has two outer electrons • Too much energy is needed to remove a third electron • This necessitates removing an electron from a shell nearer to the nucleus (to produce a Ba^{3+} ion)	1	1		2		
				Question 8 total	**9**	**7**	**0**	**16**	**2**	**2**